BASIC MECHANICAL ENGINEERING

Subject Code: 23A01201T (Part B) | UNIT – I : INTRODUCTION TO MECHANICAL ENGINEERING & ENGINEERING MATERIALS

I. Introduction – Role of Mechanical Engineering

> Mechanical Engineering is the branch of engineering that applies the principles of physics, mathematics and materials science to design, analyse, manufacture and maintain mechanical systems – machines, engines, tools, vehicles and energy systems.

- Called the "mother of all engineering branches" – broadest, oldest after civil; deals with motion, force, energy, heat and materials.
- Core areas: (i) Thermal engineering (heat engines, boilers, refrigeration), (ii) Design engineering (machine elements, CAD), (iii) Manufacturing/ production engineering, (iv) Fluid mechanics & machinery, (v) Mechatronics & robotics, (vi) Materials engineering.

❖Role in industries

- Designs & maintains machine tools, boilers, turbines, pumps, compressors, conveyors, presses.
- Production: converts raw material to finished goods – cars, appliances, electronics casings, tools.
- Plant engineering: installation, maintenance, energy audit, quality control, automation.
- Supports sectors such as power, steel, cement, textile, chemical, food processing, mining, defence.

❖Role in society

- Transport (cars, trains, ships, aircraft), energy (power plants, wind & solar machines), healthcare equipment (ventilators, prosthetics, MRI machines), air-conditioning & refrigeration (food/ medicine preservation), agriculture machinery (tractors, pumps), household appliances, sports gear.
- Improves quality of life, productivity, safety; creates jobs and drives economic growth ("Make in India").

- Responsible for sustainability – efficient engines, renewable energy devices, EVs, recycling.

> Mechanical engineer = designs the machines that move the world

2. Technologies in Different Sectors

❖(a) Energy sector

- Thermal power plants (steam turbines, boilers), gas turbines & combined cycle, hydro turbines (Pelton, Francis, Kaplan), nuclear plants.
- Renewables: wind turbines, solar thermal collectors, biomass/ biogas, geothermal; energy storage; hydrogen & fuel cells.
- Energy efficiency: heat exchangers, waste-heat recovery, cogeneration, HVAC.

❖(b) Manufacturing sector

- Casting, forging, machining, welding; CNC machines, robots, automation, CAD/CAM; 3D printing (additive manufacturing); Industry 4.0 – IoT, AI, smart factories; lean manufacturing, quality (Six Sigma).

❖(c) Automotive sector

- IC engines (petrol, diesel), transmission (gears, clutch), brakes, suspension, steering, turbochargers, fuel injection (CRDI, GDI), emission control (catalytic converter, BS-VI), Electric vehicles (EV) & hybrids, ADAS, lightweight materials (Al, composites).

❖(d) Aerospace sector

- Aircraft & rockets: aerodynamics, jet engines (turbojet, turbofan), rocket propulsion, light & strong materials (titanium, aluminium alloys, carbon fibre composites), avionics, satellite launch (ISRO – PSLV, GSLV, Chandrayaan, Gaganyaan), drones/ UAV.

❖(e) Marine sector

- Ship design & propulsion – marine diesel engines, gas turbines, propellers, rudders, hull design (naval architecture), submarines, offshore platforms, ballast & pumps, corrosion protection, LNG carriers, green shipping (hybrid/ electric ferries).

❖(f) Others

- Biomedical (implants, prosthetics), Robotics, Construction machinery, HVAC, Railways (locomotives, coaches), Defence (tanks, missiles).

3. Engineering Materials – Introduction

> Engineering materials are the substances used to make machine parts, tools and structures. Selection depends on strength, hardness, ductility, toughness, density, thermal & electrical conductivity, corrosion resistance, machinability, cost, availability.

❖Classification

Engineering materials → Metals (ferrous, non-ferrous) / Ceramics / Polymers / Composites / Smart / advanced materials

❖Mechanical properties (quick recap)

- Strength – resistance to failure under load. Elasticity – regain shape after load removal. Plasticity – permanent deformation. Ductility – drawn into wires (mild steel, copper). Malleability – hammered into sheets (gold, lead). Brittleness – fracture without deformation (cast iron, glass). Toughness – energy absorbed before fracture. Hardness – resistance to indentation/ scratch. Fatigue – failure under repeated loads. Creep – slow deformation at high T. Stiffness (Young's modulus E).

4. Metals

> Metals are crystalline, with metallic bonding, good electrical & thermal conductivity, ductility, lustre, strength.

❖4.1 Ferrous metals (iron-based)

- Contain iron as the main constituent. Iron is obtained from ores (hematite Fe_2O_3, magnetite Fe_3O_4) in a blast furnace → pig iron (3.5–4.5% C).

Material	Carbon %	Properties	Uses
Wrought iron	< 0.08	Soft, ductile, tough, rust-resistant	Chains, hooks, railings
Mild steel	0.15–0.3	Ductile, weldable, strong	Bolts, shafts, structures
Medium carbon steel	0.3–0.6	Stronger, tougher	Gears, axles, rails
High carbon steel	0.6–1.5	Very hard, brittle	Cutting tools, springs, dies
Cast iron (grey)	2–4	Brittle, good casting, damping, compressive strength	Engine blocks, machine beds, pipes
White CI / Malleable CI / SG (ductile) iron	2–4	Hard / tougher	Crankshafts, pipes

- Alloy steels: steel + Cr, Ni, Mo, V, Mn, W: stainless steel (18% Cr + 8% Ni – corrosion-proof, utensils, surgical tools), HSS (high speed steel – W, Cr, V) for cutting tools, spring steel, tool steel.
- Heat treatment (annealing, normalising, hardening, tempering, case hardening) changes properties.

❖4.2 Non-ferrous metals (no/ little iron)

Metal / Alloy	Properties	Uses
Aluminium (Al)	Light (2.7 g/cc), corrosion-resistant, good conductor, ductile	Aircraft, cans, cables, engine pistons
Copper (Cu)	Excellent electrical & thermal conductor, ductile	Wires, heat exchangers
Brass (Cu + 30–40% Zn)	Strong, corrosion-resistant, machinable	Valves, fittings, musical instruments
Bronze (Cu + Sn)	Hard, wear resistant	Bearings, bushes, gears, statues
Zinc (Zn)	Low melting, corrosion-protecting	Galvanising, die-castings
Lead (Pb)	Heavy, soft, corrosion-proof	Batteries, radiation shield
Tin (Sn)	Soft, non-toxic	Solder, tin plating
Titanium (Ti)	High strength/ weight, corrosion-proof	Aerospace, implants
Duralumin (Al+Cu+Mg)	Strong & light	Aircraft frames
Magnesium (Mg)	Lightest structural metal	Auto & aero parts
Nickel alloys (Inconel, Monel)	High-temp strength	Gas turbine blades

- Advantages of non-ferrous: light, corrosion-resistant, good conductivity, easy fabrication, non-magnetic. Disadvantages: costly, lower strength than steel.

❖Ferrous vs non-ferrous

Feature	Ferrous	Non-ferrous
Main element	Iron	Other than iron
Magnetic	Yes (mostly)	No
Corrosion	Rusts (except stainless)	Resistant
Density	High	Usually low
Cost	Cheaper	Costlier
Strength	High	Moderate

5. Ceramics

> Ceramics are inorganic, non-metallic solids made of metallic + non-metallic elements (oxides, carbides, nitrides) by shaping and high-temperature firing (sintering); ionic/ covalent bonds.

- Properties: very hard, high compressive strength, high melting point (> 2000 °C), excellent wear & corrosion resistance, electrical & thermal insulators, low density; but brittle (poor tensile strength, thermal shock).
- Types: traditional – bricks, tiles, pottery, porcelain, cement, glass; advanced/ engineering – alumina (Al_2O_3), silicon carbide (SiC), silicon nitride (Si_3N_4), zirconia (ZrO_2), tungsten carbide.
- Uses: cutting tool inserts, abrasives (grinding wheels), furnace linings/ refractories, spark plug insulators, engine/ turbine parts (thermal barrier coatings), bio-ceramics (hip joints, dental), electronic substrates, capacitors, bullet-proof armour, space shuttle tiles.

6. Polymers (brief – to complete the classification)

- Long-chain organic molecules – plastics (thermoplastics soften on heating: PVC, polyethylene, nylon, ABS; thermosets harden permanently: Bakelite, epoxy) and elastomers (rubber).
- Light, corrosion-proof, insulating, easily moulded, cheap; but low strength & temperature limit. Uses: pipes, gears (nylon), bearings, insulation, bumpers.

7. Composites

> A composite material is made by combining two or more distinct materials (matrix + reinforcement) so that the result has properties better than each constituent, with a clear interface between them.

- Matrix (binder): polymer, metal or ceramic – holds fibres, transfers load, protects them.
- Reinforcement: fibres (glass, carbon, aramid/ Kevlar), particles, flakes.

❖Classification

- By matrix: PMC (polymer matrix – most common, e.g., FRP, CFRP), MMC (metal matrix – Al-SiC), CMC (ceramic matrix – SiC-SiC).

- By reinforcement: fibre-reinforced (continuous/ short), particle-reinforced (concrete, cermet), laminar/ sandwich (plywood, honeycomb panels).

❖Properties

- High strength-to-weight and stiffness-to-weight, corrosion resistance, fatigue resistance, tailor-made (anisotropic), design flexibility, low thermal expansion.
- Drawbacks: costly, difficult to repair/ recycle, anisotropic, delamination, brittle failure.

❖Examples & applications

- Fibre glass (GFRP) – boat hulls, tanks, pipes, bathtubs, wind-turbine blades.
- Carbon fibre (CFRP) – aircraft (Boeing 787, 50% composite), F1 cars, bicycles, sports goods, satellites, prosthetics.
- Reinforced concrete, plywood, fibre cement, Kevlar (bullet-proof vests), brake pads, tungsten carbide cermet tools.

> Concrete + steel = RCC; Glass fibre + resin = GFRP; Carbon fibre + epoxy = CFRP

8. Smart Materials

> Smart (intelligent) materials are materials whose properties (shape, stiffness, colour, conductivity) change in a controlled & reversible way in response to an external stimulus such as temperature, stress, electric/ magnetic field, light, pH.

❖Types, behaviour & uses

Smart material	Stimulus → Response	Applications
Shape memory alloys (SMA) – Nitinol (Ni-Ti)	Heat → returns to original shape	Stents, eyeglass frames, actuators, aircraft wing flaps, couplings
Piezoelectric – quartz, PZT	Stress ⇄ electric voltage	Sensors, lighters, ultrasound, microphones, energy harvesting, inkjet printers
Magnetostrictive – Terfenol-D	Magnetic field → change in length	Sonar, actuators
Electro-/ magneto-rheological fluids (ER/ MR)	Field → viscosity jumps	Adaptive dampers, shock absorbers, clutches
Thermochromic / photochromic	Temp/ light → colour change	Mood rings, smart glass, spectacles
Electrochromic	Voltage → opacity	Smart windows, aircraft windows
Self-healing polymers/ concrete	Crack → heals	Coatings, infrastructure
Hydrogels, fibre-optics, shape-memory polymers	pH/ temp → swell	Drug delivery, sensors

- **Advantages:** act as sensor + actuator, adaptive, light, compact, reduce parts. **Limitations:** high cost, fatigue life, complex control.
- Used in robotics, aerospace (morphing wings), biomedical, civil (smart structures), automotive (adaptive suspension).

9. Material Selection – Summary Table

Need	Best choice
Cheap, strong structure	Mild steel
Engine block / machine bed	Grey cast iron
Light & corrosion-proof	Aluminium alloy
Electrical wire	Copper
Cutting tools	HSS, tungsten carbide, ceramics
Aircraft high strength/ weight	CFRP, titanium
Surgical/ kitchen	Stainless steel
High temperature	Ceramics, Inconel
Sensors/ actuators	Piezo, SMA

10. Important Questions

- Define mechanical engineering; explain its role in industries and society.
- Mention the technologies used in energy, automotive, aerospace, marine sectors.
- Classify engineering materials. Differentiate ferrous & non-ferrous metals with examples.
- Write about ceramics, composites and smart materials with applications.
- Compare mild steel, cast iron, brass and bronze (composition & uses).

> Remember: Mild steel 0.15–0.3% C; High carbon steel 0.6–1.5% C; Brass = Cu + Zn; Bronze = Cu + Sn

11. Extra Facts for Quick Revision

- **Steel** = iron + carbon (< 2% C); **cast iron** = iron + carbon (> 2% C).
- **Heat treatment:** *Annealing* – soften & relieve stress (slow furnace cooling); *Normalising* – refine grain (air cooling); *Hardening* – heat above critical temp, quench in water/ oil; *Tempering* – reheat after hardening to reduce brittleness; *Case hardening* – hard surface, tough core (gears).
- **Alloy** = mixture of a metal with other elements to improve properties (e.g., steel, brass, bronze, solder = Pb + Sn).
- **Corrosion** – electrochemical deterioration (rusting: $Fe + O_2 + H_2O \rightarrow$ hydrated iron oxide). **Prevention:** painting, **galvanising (Zn coat)**, plating, alloying (stainless), cathodic protection, oiling.
- **Density:** steel 7850; Al 2700; Cu 8960; Ti 4500; CFRP ~ 1600 kg/m^3.
- **Melting points:** Al 660 °C, Cu 1085 °C, Fe 1538 °C, W 3422 °C, Zn 420 °C, Pb 327 °C.
- **Stress–strain curve (mild steel):** proportional limit → elastic limit → yield point → strain hardening → ultimate stress → necking → fracture. Young's modulus **E = stress/ strain** (steel 200 GPa; Al 70 GPa; Cu 110 GPa).

12. Structure of Metals (short)

- Atoms are arranged in **crystal lattices**: **BCC** (body-centred cubic – α-iron, Cr, W; strong), **FCC** (face-centred cubic – γ-iron, Al, Cu, Ni; very ductile), **HCP** (hexagonal close packed – Zn, Mg, Ti; less ductile).
- **Grain size**: small grains → higher strength (Hall–Petch); heat treatment & cold working control grains.
- **Iron-carbon alloys phases**: ferrite (soft), **austenite** (non-magnetic, high T), **cementite** Fe_3C (very hard, brittle), **pearlite** (ferrite + cementite lamellae), **martensite** (very hard, from quenching).
- **Eutectoid steel** = 0.8% C (723 °C).

13. Testing of Materials (basics)

- **Tensile test** (UTM): specimen pulled until it breaks → gives yield strength, ultimate strength, % elongation, % reduction in area.
- **Hardness tests**: **Brinell** (HB – steel ball), **Rockwell** (HRC – diamond cone), **Vickers** (HV – diamond pyramid), **Mohs scale** (minerals).
- **Impact test**: **Charpy / Izod** – energy absorbed to break a notched bar (toughness).
- **Fatigue test** (rotating beam), **creep test**, **bend test**, **NDT** (ultrasonic, X-ray/ radiography, dye-penetrant, magnetic particle) for finding cracks without damage.

14. Materials in Real Products – Case Studies

Product / Part	Material	Why
Car body	Mild/ HSLA steel, Al	Strength, formability, weight
Engine block	Grey CI / Al alloy	Casting, damping / lightness
Piston	Aluminium alloy	Light, good conduction
Crankshaft, gears	Alloy steel, SG iron	Strength, fatigue resistance
Bearings	Bronze, babbitt	Low friction, wear
Radiator	Copper/ Al	Heat transfer
Tyres	Rubber + steel cord	Elasticity, strength
Aircraft skin	Al alloys, CFRP	Strength/ weight
Turbine blades	Ni superalloy + ceramic coat	High temperature
Ship hull	Mild steel, FRP (small boats)	Strength, cost
Cutting tool	HSS, carbide, ceramic	Hardness at high temperature
Electrical wire	Copper, Al	Conductivity
Medical implant	Ti, stainless steel, bioceramic	Bio-compatibility

15. Sector Technology Snapshot

- **Energy:** efficiency of a steam plant ≈ 35–40%, combined cycle gas plant ≈ 55–60%; hydro ≈ 90%; wind turbine $C_p \leq 0.59$.
- **Automotive:** typical petrol engine efficiency 25–30%, diesel 35–40%; EV motor ≈ 90%; BS-VI emission standard since 2020.
- **Aerospace:** lift = $\frac{1}{2} \rho V^2 S C_L$; thrust from Newton's 3rd law (jet/ rocket); ISRO's Mars/ Moon missions use cryogenic engines.
- **Marine:** buoyancy (Archimedes' principle) – ship floats when weight of water displaced = weight of ship; propellers efficiency 60–70%.
- **Manufacturing:** automation + robotics + AI → Industry 4.0; additive manufacturing reduces waste.

16. Key Definitions (for 2-mark answers)

- **Ferrous metal:** metal containing iron. **Non-ferrous:** without iron (Al, Cu, Zn...).
- **Alloy:** metal mixed with other elements. **Ceramic:** inorganic non-metallic, sintered.

- **Composite:** matrix + reinforcement. **Smart material:** responds to external stimulus.
- **Piezoelectric effect:** pressure ⇄ voltage. **Shape memory effect:** deformed material returns on heating.
- **Machinability:** ease of cutting. **Weldability:** ease of joining by welding. **Castability:** ease of casting.
- **Anisotropic:** properties differ with direction (composites). **Isotropic:** same in all directions (metals).

17. Short Quiz (Practice)

- Which metal is called the "lightest structural metal"? → **Magnesium.**
- Brass is an alloy of? → **Copper + Zinc.** Bronze? → **Copper + Tin.**
- Which cast iron is brittle but has good damping? → **Grey cast iron.**
- Stainless steel main alloying elements? → **Chromium (≥ 10.5%) + Nickel.**
- Which ceramic is used for spark plug insulators? → **Alumina.**
- Nitinol is an example of? → **Shape memory alloy.**
- CFRP stands for? → **Carbon fibre reinforced polymer.**
- Which property allows drawing into wires? → **Ductility.**